

Holes and Vertical Asymptotes

Answer Key

Algebra 2

Quick Answers

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|---|--|
| 1. $\frac{x+3}{x+2}$
2. (a) zeros of the original denominator = -2 ,
3, —
3. $\frac{6}{5}$
4. (a) simplified form = $\frac{2(x+2)}{x+3}$, (b) values
excluded from the domain = $-3, 2$ | 5. (a) the correct height of the hole = 4 , —
6. (a) simplified form = $\frac{x-3}{x-2}$, (b) the vertical
asymptote = 2 , —
7. (a) simplified form = x , (b) x-values of the
holes = $-1, 1$, —
8. (a) simplified form = $\frac{x+5}{x+1}$, (b) height of the
hole = 2 , (c) the vertical asymptote = -1 |
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What is verified

13 of 17 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 2, 5, 6, 7. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-APR.D.6 – problems 1, 4, 6, 7, 8

HSF-IF.C.7 – problems 2, 3, 5, 6, 7, 8

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

3. If they got 0: put $x = 3$ straight into the original expression and reported the height as 0, because the numerator is zero there
5. If they got 0: read the height of the hole as zero because the numerator is zero at $x = 2$
6. If they got 2, -2 : treated every zero of the original denominator as a vertical asymptote, keeping the cancelled factor's zero
8. If they got 0: cancelled correctly but then put $x = 3$ into the original numerator and reported the height as 0
8. If they got $-1, 3$: kept the cancelled factor's zero and reported an extra asymptote at $x = 3$
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1. Write $\frac{x^2 - 9}{x^2 - x - 6}$ in fully simplified form.

Solution: Factor both parts, then cancel the shared factor:

$$\frac{x^2 - 9}{x^2 - x - 6} = \frac{(x - 3)(x + 3)}{(x - 3)(x + 2)}.$$

The factor $(x - 3)$ is common to top and bottom, so it cancels.

$$\frac{x + 3}{x + 2}$$

2. $f(x) = \frac{x^2 - 9}{x^2 - x - 6}$. (a) Zeros of the original denominator. (b) Which is a hole, which is a vertical asymptote, and why.

Solution (a): $x^2 - x - 6 = (x - 3)(x + 2)$, which is zero when $x = 3$ or $x = -2$.

$x = -2 \text{ or } x = 3$

Solution (b): *Instructor-judged.* After factoring, $f(x) = \frac{(x - 3)(x + 3)}{(x - 3)(x + 2)}$. The factor $(x - 3)$ appears above and below and cancels, so $x = 3$ is a **hole** — the simplified function is perfectly well behaved there, the original just has no value assigned. The factor $(x + 2)$ survives the cancelling, so as $x \rightarrow -2$ the denominator goes to zero while the numerator does not, and the values run off without bound: $x = -2$ is a **vertical asymptote**. Full credit needs both labels and the cancelling reason.

3. Coordinates of the hole in the graph of $f(x) = \frac{x^2 - 9}{x^2 - x - 6}$.

Solution: The hole sits at the x -value of the cancelled factor, $x = 3$. Its height is the value the *simplified* function takes there:

$$\left. \frac{x + 3}{x + 2} \right|_{x=3} = \frac{6}{5}.$$

So the graph is a smooth curve with one point missing, at

$\left(3, \frac{6}{5}\right)$

4. $g(x) = \frac{2x^2 - 8}{x^2 + x - 6}$. (a) Simplify. (b) Values excluded from the domain.

Solution (a): Take out the common factor 2 on top, then factor both quadratics:

$$\frac{2x^2 - 8}{x^2 + x - 6} = \frac{2(x - 2)(x + 2)}{(x + 3)(x - 2)}.$$

Cancelling $(x - 2)$ leaves

$$\frac{2(x + 2)}{x + 3}$$

Solution (b): The domain is set by the *original* denominator, before any cancelling: $(x + 3)(x - 2) = 0$ at $x = -3$ and $x = 2$. Cancelling removes the factor from the formula, not the restriction from the domain.

$x = -3 \text{ or } x = 2$

5. Student work on the hole of $h(x) = \frac{x^2 - 4}{x - 2}$: “the top is 0 at $x = 2$, so the hole is at height 0.” (a) Correct height. (b) The error.

Solution (a): $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for every $x \neq 2$. The height of the hole is the value of that cancelled form at $x = 2$, namely $2 + 2$. 4

Solution (b): *Instructor-judged.* The error is substituting into the original quotient. At $x = 2$ the numerator and the denominator are *both* zero, so the expression $\frac{0}{0}$ says nothing at all — it is not 0. The shared factor $(x - 2)$ has to be cancelled first; the height of the hole is then read from $x + 2$. A full-credit answer says a zero numerator does not force the value to be zero when the denominator vanishes at the same place.

6. Student work on $k(x) = \frac{x^2 - x - 6}{x^2 - 4}$: “the bottom is zero at $x = 2$ and $x = -2$, so both are vertical asymptotes.” (a) Simplify. (b) The only vertical asymptote. (c) The error.

Solution (a): $\frac{x^2 - x - 6}{x^2 - 4} = \frac{(x - 3)(x + 2)}{(x - 2)(x + 2)}$, and $(x + 2)$ cancels.

$$\frac{x - 3}{x - 2}$$

Solution (b): Only the denominator left after cancelling can produce an asymptote, and $x - 2 = 0$ gives x = 2

Solution (c): *Instructor-judged.* The student used the original denominator instead of the cancelled one. Because $(x + 2)$ divides out, the graph does not run off to infinity at $x = -2$; it simply has one missing point there, a hole at $x = -2$. Vertical asymptotes come from the factors that survive cancelling. A full-credit answer names $x = -2$ as a hole and states that rule.

7. $m(x) = \frac{x^3 - x}{x^2 - 1}$. (a) Simplify. (b) x -values of the holes. (c) Number of vertical asymptotes, with a reason.

Solution (a): Factor completely:

$$\frac{x^3 - x}{x^2 - 1} = \frac{x(x - 1)(x + 1)}{(x - 1)(x + 1)}.$$

Both denominator factors cancel.

$m(x) = x$

Solution (b): The cancelled factors were $(x - 1)$ and $(x + 1)$, so the graph is the line $y = x$ with two points removed. x = -1 or x = 1

Solution (c): *Instructor-judged.* There are **none**. Every factor of the denominator cancelled, so nothing is left to drive the function to infinity; the graph is the straight line $y = x$ with holes at $(-1, -1)$ and $(1, 1)$. A full-credit answer says zero asymptotes *and* explains that all denominator factors cancelled.

8. $n(x) = \frac{x^2 + 2x - 15}{x^2 - 2x - 3}$. (a) Simplify. (b) Height of the hole. (c) The vertical asymptote.

Solution (a): $\frac{x^2 + 2x - 15}{x^2 - 2x - 3} = \frac{(x + 5)(x - 3)}{(x - 3)(x + 1)}$, and $(x - 3)$ cancels.

$$\frac{x + 5}{x + 1}$$

Solution (b): The cancelled factor puts the hole at $x = 3$; its height comes from the simplified form: $\frac{3 + 5}{3 + 1} = \frac{8}{4}$.

$$2$$

Solution (c): The surviving denominator factor is $(x + 1)$, zero at

$$x = -1$$